4 Weeks 3-4: Potential Outcomes and Randomization-Based Inference

4.1 Science and Assignment

A central conceptual contribution of this line of statistical inference is the strict separation between two components of an experiment:

1. **The science:** the collection of potential outcomes

$$\mathcal{S} = \{Y_i(1), Y_i(0) : i = 1, \ldots, N\}.$$

2. **The assignment:** the randomized mechanism that determines which potential outcome is observed for each unit.

The science describes how the world *would respond* to different treatments. The assignment describes how nature (or the experimenter) chooses which response is revealed.

This distinction is not merely philosophical. All probability statements in randomized inference are conditional on the science and arise exclusively from the assignment. Potential outcomes themselves are treated as fixed, unknown quantities.

This framing sharply contrasts with statistical models that posit probability laws for outcomes themselves.

4.2 Potential Outcomes Revisited: Treatment as a Cause

Recall that for each unit i, we define two potential outcomes:

$$Y_i(1), \quad Y_i(0).$$

These are not random variables in our design based framework. They are fixed attributes of unit i, just like height or age.

The causal effect of treatment on unit i is

$$\tau_i = Y_i(1) - Y_i(0).$$

Causality is therefore defined at the unit level, prior to any averaging or inference. Randomization does not *define* causality; it enables us to reason about it.

4.3 Observed Data as a Function of Assignment

Let $\mathbf{Z} = (Z_1, \ldots, Z_N)$ denote the treatment assignment vector. The observed outcome for unit i is

$$Y_i^{\text{obs}} = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

Thus, the observed data are a deterministic function of:

- the fixed science $\mathcal{S}$,
- the realized assignment $\mathbf{Z}$.

All uncertainty in inference arises because $\mathbf{Z}$ could have been different.

4.4 Randomization Distributions: The Central Object

Fix the science $\mathcal{S}$. Let G denote the set of all treatment assignments allowed by the design.

For any statistic $T(\mathbf{Z}, \mathcal{S})$, the *randomization distribution* is the distribution induced by evaluating T over all $\mathbf{Z} \in G$, weighted by the known assignment probabilities.

Formally,

$$\Pr\{T \leq t \mid \mathcal{S}\} = \sum_{\mathbf{z} \in G} \mathbb{I}\{T(\mathbf{z}, \mathcal{S}) \leq t\} \Pr(\mathbf{Z} = \mathbf{z}).$$

Key Insight. This is a probability distribution over *assignments*, not outcomes. The data-generating mechanism is the randomization itself.

4.5 Fisher's Sharp Null Hypothesis

Fisher proposed testing the hypothesis

$$H_0 : Y_i(1) = Y_i(0) \quad \forall i.$$

This null hypothesis is called *sharp* because it uniquely determines all missing potential outcomes.

Under H_0, the science $\mathcal{S}$ is fully known:

$$Y_i(1) = Y_i(0) = Y_i^{\text{obs}}.$$

Consequently, for every possible assignment $\mathbf{z} \in G$, the statistic $T(\mathbf{z}, \mathcal{S})$ can be computed exactly.

4.6 Exact Randomization Tests

Given an observed statistic T_{obs}, the randomization p-value is

$$p = \Pr\left(|T(\mathbf{Z}, \mathcal{S})| \geq |T_{\text{obs}}| \mid H_0\right).$$

This probability is computed with respect to the randomization distribution.

Crucial Properties.

- The test is exact in finite samples.
- No distributional assumptions on outcomes are required.
- Validity depends only on the correctness of the assignment mechanism.

4.7 Why Randomization Inference Is Not Permutation Testing

Although randomization tests resemble permutation tests computationally, we emphasize that they are conceptually distinct.

Permutation tests rely on exchangeability assumptions about the joint distribution of outcomes. Randomization tests rely only on the physical act of randomization.

Thus, randomization inference remains valid even when outcome distributions are heteroskedastic, skewed, or heavy-tailed.

4.8 Constant Additive Treatment Effects

Fisher's sharp null is often implausible. A weaker, yet still highly structured model is the *constant additive treatment effect* assumption:

$$Y_i(1) = Y_i(0) + \tau, \quad \forall i.$$

This is a causal model, not merely a statistical convenience. It asserts that treatment shifts outcomes by the same amount for every unit.

4.9 Testing Nonzero Effects via Outcome Adjustment

To test

Under the constant additive treatment effect assumption, consider testing the hypothesis

$$H_0 : \tau = \tau_0.$$

This is still a sharp null hypothesis as we can impute the counterfactuals. To see that, define adjusted outcomes

$$Y_i^{(\tau_0)} = Y_i^{\text{obs}} - Z_i \tau_0.$$

If H_0 is true, then

$$Y_i^{(\tau_0)} = Y_i(0),$$

and the adjusted data satisfy Fisher's sharp null of no treatment effect.

Therefore, randomization tests for $H_0 : \tau = 0$ applied to $Y^{(\tau_0)}$ are valid tests of $H_0 : \tau = \tau_0$.

4.9.1 Randomization Test at Level α

Let $T(\mathbf{Z}, \mathbf{Y})$ be a test statistic. For each τ_0, compute the randomization p-value

$$p(\tau_0) = \Pr\Big(|T(\mathbf{Z}, \mathbf{Y}^{(\tau_0)})| \geq |T(\mathbf{Z}^{\text{obs}}, \mathbf{Y}^{(\tau_0)})| \;\Big|\; H_0(\tau_0) \Big).$$

Reject $H_0(\tau_0)$ if $p(\tau_0) \leq \alpha$.

4.9.2 Inversion and Confidence Sets

The $(1-\alpha)$ confidence set for τ is defined as

$$\mathcal{C}_{1-\alpha} = \{\tau_0 \in \mathbb{R} : p(\tau_0) > \alpha\}.$$

Theorem 4. *Under the constant additive treatment effect model, $\mathcal{C}_{1-\alpha}$ has exact finite-sample coverage:*

$$\Pr(\tau \in \mathcal{C}_{1-\alpha}) \geq 1 - \alpha.$$

Proof. If the true treatment effect is τ, then $H_0(\tau)$ is true. By validity of the randomization test,

$$\Pr(p(\tau) \leq \alpha) \leq \alpha.$$

Equivalently,

$$\Pr(p(\tau) > \alpha) \geq 1 - \alpha,$$

which implies $\Pr(\tau \in \mathcal{C}_{1-\alpha}) \geq 1 - \alpha$. □

4.9.3 Structure of the Confidence Interval

For many statistics (e.g. difference in means), $T(\mathbf{Z}, \mathbf{Y}^{(\tau_0)})$ is monotone in τ_0. In such cases, $\mathcal{C}_{1-\alpha}$ is an interval of the form

$$[\tau_L, \tau_U].$$

This monotonicity allows confidence intervals to be computed efficiently by searching over τ_0 values.

4.9.4 Interpretation

Randomization-based confidence intervals:

- are exact in finite samples,
- do not rely on asymptotic normality,
- depend explicitly on the chosen test statistic,
- reflect the design of the experiment rather than outcome modeling assumptions.

4.10 Relationship to Neymanian Inference

Neyman's framework targets the average treatment effect without assuming constant effects. Fisher's framework enables exact testing under sharp hypotheses.

Rosenbaum emphasizes that these approaches are complementary:

- Neyman focuses on estimation and repeated-sampling properties.
- Fisher focuses on testing and exact inference.

The constant additive effect model provides a bridge between the two.

4.11 Why This Framework Matters

Randomization-based inference clarifies:

- what is assumed,
- what is tested,
- where probability enters,
- and why causal claims are justified.

It provides a benchmark against which all observational methods should be judged.

4.12 Choice of Test Statistics in Randomization-Based Inference

4.12.1 Why Test Statistics Matter

In randomization-based inference, the assignment mechanism determines the source of randomness, but the *test statistic* determines how evidence against a null hypothesis is summarized.

A test statistic is a function

$$T(\mathbf{Z}, \mathbf{Y})$$

that maps the observed assignment and outcomes to a real number. While the randomization distribution governs how T varies across assignments, the statistic itself reflects substantive choices about what constitutes evidence of a treatment effect.

Thus, the choice of test statistic is not purely technical; it encodes scientific judgment about the expected nature of treatment effects.

4.12.2 Test Statistics as Functions on the Assignment Space

Let $\mathcal{S}$ denote the fixed science (potential outcomes) and let G denote the set of all treatment assignments allowed by the design.

A test statistic is a function

$$T : G \times \mathcal{S} \to \mathbb{R}.$$

Under a sharp null hypothesis, $\mathcal{S}$ is known, and $T(\mathbf{Z}, \mathcal{S})$ is a random variable solely through $\mathbf{Z}$.

4.12.3 Difference-in-Means Statistic

Define

$$T_{\mathrm{DM}}(\mathbf{Z}, \mathbf{Y}) = \frac{1}{n_1} \sum_{i:Z_i=1} Y_i - \frac{1}{n_0} \sum_{i:Z_i=0} Y_i.$$

Under constant additive treatment effects,

$$\mathrm{E}[T_{\mathrm{DM}} \mid \mathcal{S}] = \tau,$$

and the statistic is a natural choice for a test statistic.[1]

4.12.4 Linear Rank Statistics

Let R_i denote the rank of Y_i among all units. A general linear rank statistic is

$$T_{\mathrm{rank}} = \sum_{i=1}^{N} Z_i a(R_i),$$

where $a(\cdot)$ is a score function (e.g. Wilcoxon, normal scores).

Key Property. The null randomization distribution of T_{rank} depends only on the ranks, not the outcome values themselves.

This yields robustness to heavy tails, skewness, and outliers.

[1] In fact this statistic is optimal (in a Pitman efficiency sense) when outcomes are homoskedastic and treatment effects are homogeneous.

4.12.5 Attributable Effect Statistic

This statistic is particularly relevant whe the outcome is binary. Define the attributable effect

$$A(\mathbf{Z}, \mathcal{S}) = \sum_{i=1}^{N} Z_i\big(Y_i(1) - Y_i(0)\big).$$

Under Fisher's sharp null, $A = 0$ for all assignments. Under alternatives, A measures the aggregate effect attributable to treatment among treated units.

This statistic is particularly useful when scientific interest focuses on counts of events caused by treatment.

4.12.6 Design Compatibility

If the experiment uses blocking or pairing, the statistic should respect the design. For example, in paired designs,

$$T_{\text{pair}} = \sum_{b=1}^{B} (Y_{bT} - Y_{bC}),$$

where (bT, bC) index treated and control units within block b.

Ignoring the design can lead to loss of power even though validity is preserved.

4.12.7 Summary

Test statistics play a dual role:

1. They summarize evidence against a null hypothesis.
2. They encode scientific beliefs about plausible treatment effects.

Randomization-based inference makes this role explicit and transparent.

4.13 A/B Testing in Mobile App Development

In mobile A/B testing, users are randomly assigned to different versions of an app. Randomization ensures that all user characteristics—observed and unobserved—are balanced in expectation, making causal interpretation straightforward even in large, heterogeneous populations.

Example: Notification Timing.

- Units: users
- Treatment $Z = 1$: notification at 8am
- Control $Z = 0$: notification at 6pm
- Outcome: app opened within 24 hours

Users are randomly assigned with known probabilities. All user characteristics (device type, engagement history, preferences) are rendered irrelevant for identification.

This is a textbook completely randomized experiment.

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4.14 Covariates in Randomized Experiments

In randomized experiments:

- Covariates are *not required* for unbiasedness
- Covariates can improve precision
- Covariates motivate blocking and stratification

Rosenbaum emphasizes:

> Adjustment for covariates refines an experiment; it does not rescue a poor design.

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4.15 Why Randomized Experiments Are the Gold Standard

Randomization-based inference reverses the usual modeling logic:

Model-based inference	**Randomization-based inference**
Probability from outcome model	Probability from design
Assumptions on $Y \mid X, Z$	No assumptions on outcomes
Asymptotics often required	Exact finite-sample inference

As Rosenbaum emphasizes, randomization inference provides:

> *causal conclusions without modeling assumptions*, relying only on the known assignment mechanism.

This is why randomized experiments are the **gold standard**: they convert causal questions into probability calculations over known randomization distributions, rather than unverifiable assumptions about nature.

Randomized experiments are the gold standard because:

1. Identification rests on known, transparent mechanisms
2. Inference is valid in finite samples
3. No untestable assumptions about outcomes are required
4. Sensitivity to hidden bias is zero by design

Observational studies attempt to approximate these properties. Randomization achieves them exactly.

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4.16 Limitations

Randomized experiments may still face:

- Noncompliance
- Attrition
- Interference between units
- Ethical or logistical constraints

Homeworks

Problem 5 (Numerical A/B Testing Example)**.** *A mobile app company runs an A/B test on $N = 10$ users. Five users are randomly assigned to the treatment (new interface) and five to control (old interface). The observed outcomes (number of sessions in a week) are:*

Treatment	8	6	7	9	5
Control	4	6	5	3	4

1. *Compute the observed difference in means.*
2. *Enumerate all possible treatment assignments consistent with the design.*
3. *Compute the randomization distribution of the difference-in-means statistic under the sharp null hypothesis of no treatment effect.*
4. *Compute the exact p-value.*

Problem 6 (Finite Population Perspective)**.** *Consider a randomized experiment with fixed potential outcomes $\{Y_i(1), Y_i(0)\}_{i=1}^N$.*

1. *Explain why, in the Neymanian framework, the only source of randomness is the treatment assignment.*
2. *Show that the sample average treatment effect*

$$\tau = \frac{1}{N}\sum_{i=1}^{N}(Y_i(1) - Y_i(0))$$

   is a fixed (non-random) quantity.

3. *Contrast this perspective with a superpopulation model.*

Problem 7 (Randomization-Based vs Model-Based Inference)**.** *Suppose an experimenter analyzes data from a completely randomized experiment using both:*

- *a randomization-based t-test, and*
- *a linear regression with homoskedastic errors.*

1. *Explain why these two procedures often yield similar numerical results.*
2. *Describe a setting in which the regression-based standard errors may be misleading.*
3. *Explain why Rosenbaum argues that randomization-based inference is conceptually primary.*